

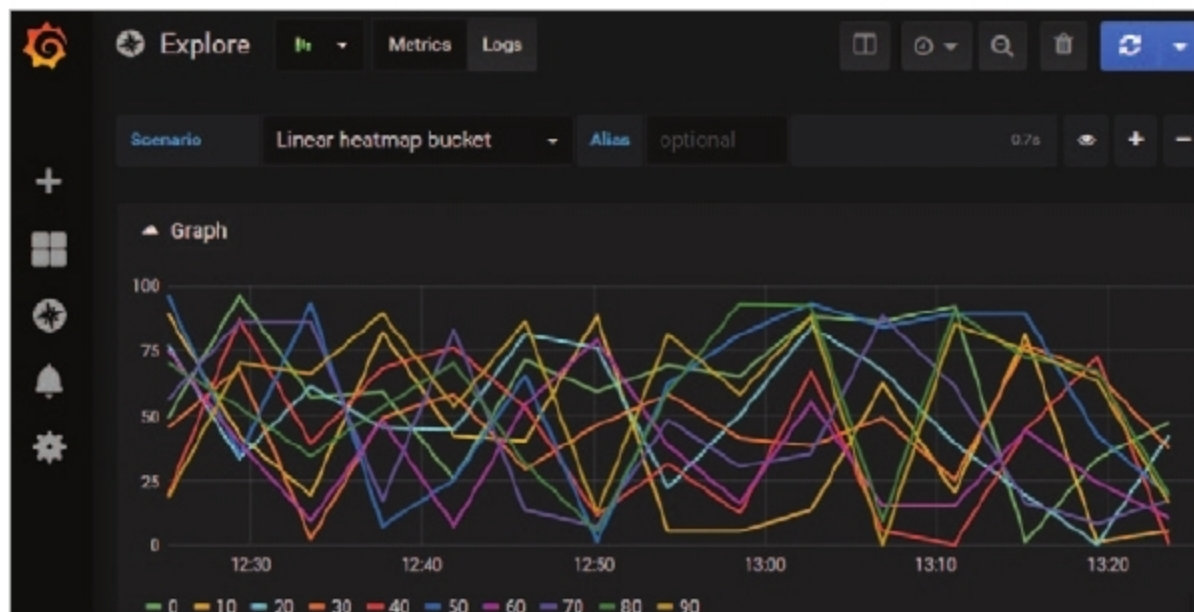


Figure 8: Visualisation with Heatmap Bucket

science projects associated with different applications (Figure 6).

Using the Grafana tool, the plotting and visualisation tasks can be done with multiple types of graphs and charts, ensuring that data analytics is highly accurate (Figure 7).

Besides the traditional bar graphs, line graphs and bubble plots,

a number of multi-dimensional visualisation based plots are available in Grafana (Figure 8).

Scope for research and development

For research projects associated with data science and predictive knowledge discovery, there are numerous software

tools and frameworks available that have their own key features. Grafana is a high performance tool that is used by data scientists and researchers for multi-dimensional plotting and visualisation, which are very important to complete any real-time data science project. Researchers and practitioners can use Grafana Desktop or Grafana Cloud for the tasks associated with visualisation of scientific data and the results obtained from the data science algorithms, so that accurate outcomes can be presented. **END** 🐧

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Here again the choice of language often depends on personal priorities. But very often, the preferred language for code obfuscation is Perl, which is famous for its useful one-liners and notorious for producing unreadable code. Code obfuscation competitions are regularly held for Perl and C programmers. The most famous code obfuscation competition called International Obfuscated C Code Contest (IOCCC) is held for C programmers. You can find the official website of IOCCC at <http://www.de.ioccc.org/>. I urge you to go through the winning entries of IOCCC. I was really fascinated and felt humbled by the amazing code produced by the winners.

Now it is time for us to wind up our discussion. So far, we have discussed code golfing and code obfuscation as the hobbies of probably just a few geeks. Such activities are very important because many popular programming

Figure 5: Output of the modified program *kerala2.c*

contests have code golfing and code obfuscation rounds. If you participate in these contests regularly, you will have a tighter grasp over the minute features of programming languages. Moreover, code obfuscation makes reverse engineering a lot more difficult. I am sure you are aware that having the ability to write code that is difficult to reverse engineer is a skill that is highly valued in programming circles. So, I assure you, you will be aptly rewarded for the time spent on code golfing and code obfuscation competitions. **END** 🐧

By: Deepu Benson

The author is a free software enthusiast whose area of interest is theoretical computer science. The open source tools of his choice include ns-2 and ns-3. He maintains a technical blog at www.computingforbeginners.blogspot.in.